Using Smartbands, Pupillometry and Body Motion to Detect Discomfort in Automated Driving

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Abstract

As technological advances lead to rapid progress in driving automation, human-machine interaction (HMI) issues such as comfort in automated driving gain increasing attention. The research project KomfoPilot at Chemnitz University of Technology aims to assess discomfort in automated driving using physiological parameters from commercially available smartbands, pupillometry and body motion. Detected discomfort should subsequently be used to adapt driving parameters as well as information presentation and prevent potentially safety-critical take-over situations. In an empirical driving simulator study, 40 participants from 25 years to 84 years old experienced two highly automated drives with three potentially critical and discomfort-inducing approaching situations in each trip. The ego car drove in a highly automated mode at 100 km/h and approached a truck driving ahead with a constant speed of 80 km/h. Automated braking started very late at a distance of 9 m, reaching a minimum of 4.2 m. Perceived discomfort was assessed continuously using a handset control. Physiological parameters were measured by the smartband Microsoft Band 2 and included heart rate (HR), heart rate variability (HRV) and skin conductance level (SCL). Eye tracking glasses recorded pupil diameter and eye blink frequency; body motion was captured by a motion tracking system and a seat pressure mat. Trends of all parameters were analyzed 10 s before, during and 10 s after reported discomfort to check for overall parameter relevance, direction and strength of effects; timings of increase/decrease; variability as well as filtering, standardization and artifact removal strategies to increase the signal-to-noise ratio. Results showed a reduced eye blink rate during discomfort as well as pupil dilation, also after correcting for ambient light influence. Contrary to expectations, HR decreased significantly during discomfort periods, whereas HRV diminished as expected. No effects could be observed for SCL. Body motion showed the expected pushback movement during the close approach situation. Overall, besides SCL, all other parameters showed changes associated with discomfort indicated by the handset control. The results serve as a basis for designing and configuring a real-time discomfort detection algorithm that will be implemented in the driving simulator and validated in subsequent studies.